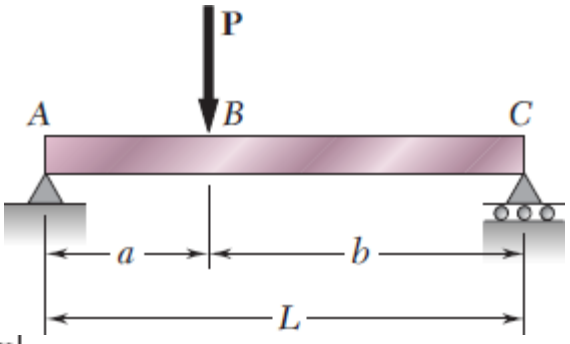
Question 1.

Find equations for Shear and bending moment, and draw the shear and bending-moment diagrams for the beam and loading shown.



Use P = 4kN, a = 2m, b=3m

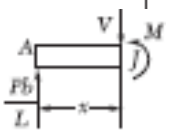
From A to B $\quad 0 < x < a$

$$+\uparrow \Sigma F_y = 0 \qquad \frac{Pb}{L} - V = 0$$

$$V = \frac{Pb}{L}$$

$$\circlearrowleft \Sigma M_J = 0 \qquad M - \frac{Pb}{L}x = 0$$

$$M = \frac{Pbx}{L}$$

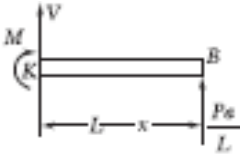


Substituting: $v = 2.4\ kN \qquad M = 2.4x\ kNm$

From B to C $\quad a < x < L$

$$+\uparrow \Sigma F_y = 0 \qquad V + \frac{Pa}{L} = 0$$

$$V = -\frac{Pa}{L}$$



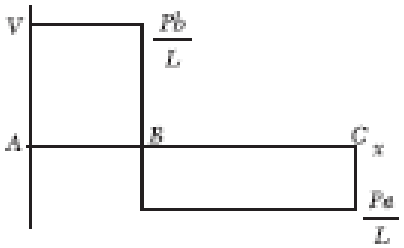
Substituting: $v = 1.6\ kN$

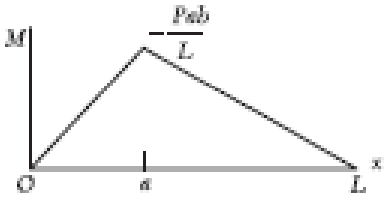
$$\circlearrowleft \Sigma M_K = 0 \qquad -M + \frac{Pa}{L}(L - x) = 0$$

$$M = \frac{Pa(L - x)}{L}$$

Substituting: $M = 8 - 1.6x\ kNm$

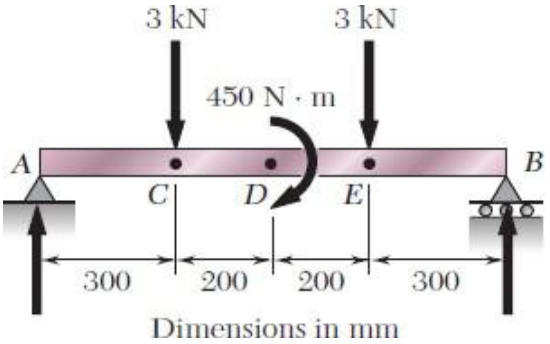
At section B $\qquad M = \frac{Pab}{L}$





Question 2.

Derive equations for the shear force and the bending moment and draw the shear and bending-moment diagrams for the beam and loading shown, and determine the maximum absolute value (a) of the shear, (a) of the bending moment.



SOLUTION

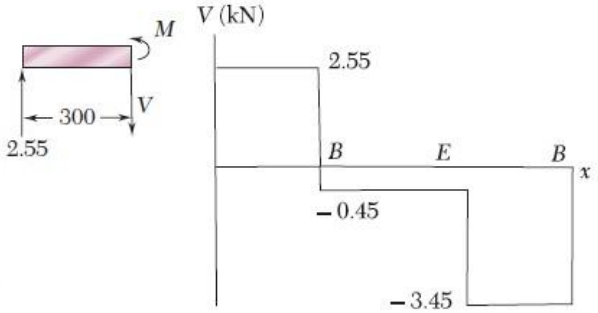
Dimensions in mm

$\circlearrowleft^{+} \Sigma M_B = 0 \qquad (700)(3) - 450 + (300)(3) - 1000A = 0$

$A = 2.55 \text{ kN} \uparrow$

$\circlearrowleft^{+} \Sigma M_A = 0 \qquad -(300)(3) - 450 - (700)(3) + 1000B = 0$

$B = 3.45 \text{ kN} \uparrow$



At A $\quad V = 2.55 \text{ kN} \qquad M = 0$

A to C $\quad V = 2.55 \text{ kN}$

At C $\qquad \circlearrowleft^{+} \Sigma M_C = 0$

$-(300)(2.55) + M = 0$

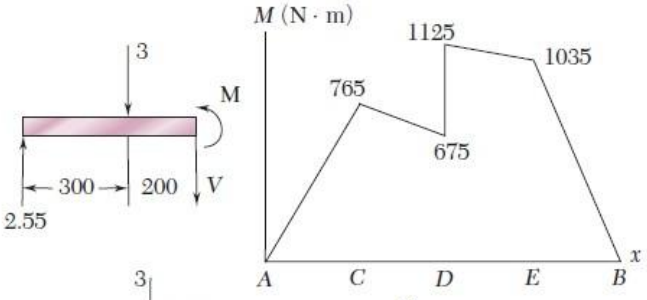
$M = 7.65 \text{ N} \cdot \text{m}$

C to E $\quad V = -0.45 \text{ N} \cdot \text{m}$

At D^- $\quad \circlearrowleft^{+} \Sigma M_D = 0$

$-(500)(2.55) + (200)(3) + M = 0$

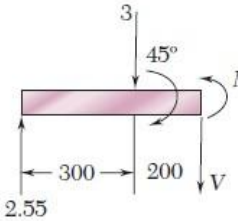
$M = 675 \text{ N} \cdot \text{m}$



At D^+ $\qquad \circlearrowleft^{+} \Sigma M_D = 0$

$-(500)(2.55) + (200)(3) - 450 + M = 0$

$M = 1125 \text{ N} \cdot \text{m}$

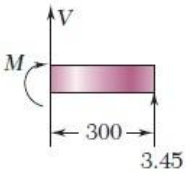


E to B $\quad V = -3.45 \text{ kN}$

At E $\qquad \circlearrowleft^{+} \Sigma M_E = 0$

$-M + (300)(3.45) = 0$

$M = 1035 \text{ N} \cdot \text{m}$



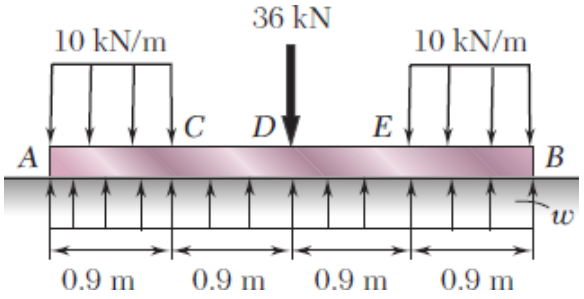
At B $\quad V = 3.45 \text{ N} \cdot \text{m}, \qquad M = 0$

Maximum $|V| = 3.45 \text{ kN}$

Maximum $|M| = 1125 \text{ N} \cdot \text{m}$

Question 3.

Assume that the upward reaction of the ground is be uniformly distributed. Derive equations for the shear force and the bending moment and draw the shear and bending-moment diagrams for the beam AB and determine the maximum value (a) of the shear, (b) of the bending moment.



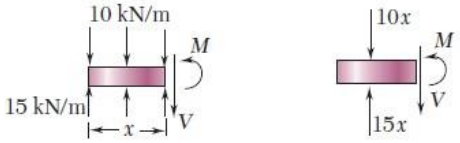
SOLUTION

Over whole beam $+\uparrow \Sigma F_y = 0$

$$3.6w - (0.9)(10) - 36 - (0.9)(10) = 0$$

$$w = 15 \text{ kN/m}$$

A to C $\quad 0 < x < 0.9$ m



$$+\uparrow \Sigma F_y = 0 \qquad 15x - 10x - V = 0$$

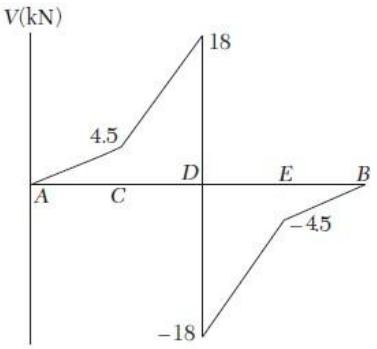
$$V = 5x$$

$$\circlearrowleft \Sigma M_J = 0 \qquad -(15x)\frac{x}{2} + (10x)\frac{x}{2} + M = 0$$

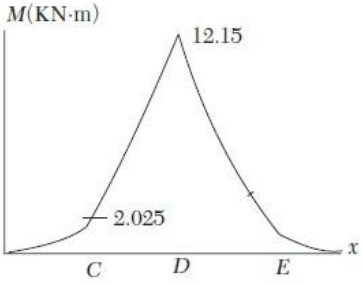
$$M = 2.5x^2$$

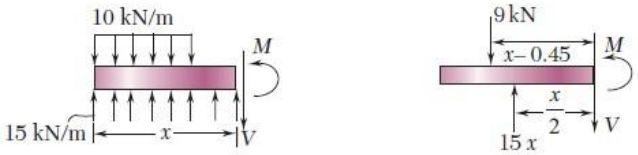
At $x = C$ $\quad V = 4.5$ kN

$M = 2.025$ kN·m



C to D $\quad 0.9 \text{ m} < x < 1.8$ m





$$+\uparrow \Sigma F_y = 0 \qquad 15x - 9 - V = 0$$

$$V = 15x - 9$$

$$\circlearrowleft \Sigma M_J = 0 \qquad -(15x)\left(\frac{x}{2}\right) + 9(x - 0.45) + M = 0$$

$$M = 7.5x^2 - 9x + 4.05 = 0$$

At D^- $\quad V = 18$ kN

$M = 12.15$ kN·m

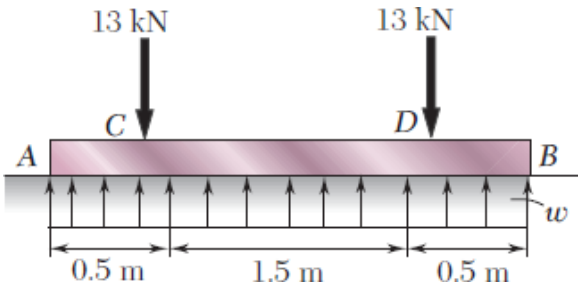
D to B Use symmetry to calculate the shear bending moment.

(a) Maximum $|V| = 18$ kN ◄

(b) Maximum $|M| = 12.15$ kN·m ◄

Question 4.

Assume that the upward reaction of the ground is be uniformly distributed. Derive equations for the shear force and the bending moment and draw the shear and bending-moment diagrams for the beam AB and determine the maximum value (a) of the shear, (b) of the bending moment.



SOLUTION

Over the whole beam

$$+\uparrow \Sigma F_y = 0 \qquad 2.5w - 13 - 13 = 0$$

$$w = 10.4 \text{ kN/m}$$

A to C $\qquad 0 < x < 0.5 \text{ m}$

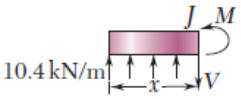
$$+\uparrow \Sigma F_y = 0 \qquad 10.4x - V = 0$$

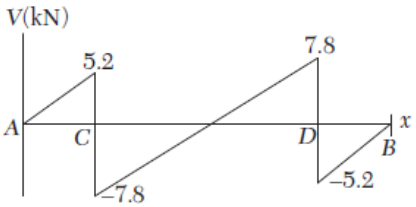
$$V = 10.4x$$

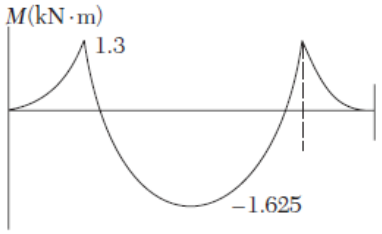
$$\circlearrowleft \Sigma M_J = 0$$

$$-(10.4x)\left(\frac{x}{2}\right) + M = 0$$

$$M = 5.2x^2$$







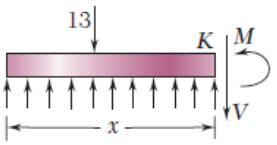
At C^- $\qquad V = 5.2 \text{ kN}, \qquad M = 1.3 \text{ kN} \cdot \text{m}$

C to D $\qquad 0.5 \text{ m} < x < 2.0 \text{ m}$

$$+\uparrow \Sigma F_y = 0$$

$$10.4x - 13 - V = 0$$

$$V = 10.4x - 13$$



$$\circlearrowleft \Sigma M_K = 0 \qquad -(10.4x)\left(\frac{x}{2}\right) + 13(x - 0.5) + M = 0$$

$$M = 5.2x^2 - 13x + 6.5$$

At the center of the beam $\qquad x = 1.25 \text{ m}$

$$V = 0, \qquad M = -1.625 \text{ kN} \cdot \text{m}$$

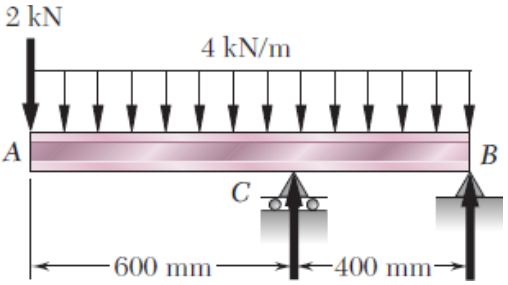
At C^+ $\qquad V = -7.8 \text{ kN} \qquad M = 1.3 \text{ kN} \cdot \text{m}$

(a) Maximum $|V| = 7.8$ kN ◄

(b) Maximum $|M| = 1.625$ kN·m ◄

Question 5.

Derive equations for the shear force and the bending moment and draw the shear and bending-moment diagrams for the beam and loading shown and determine the maximum bending moment.





SOLUTION

$$\circlearrowleft \Sigma M_C = 0$$

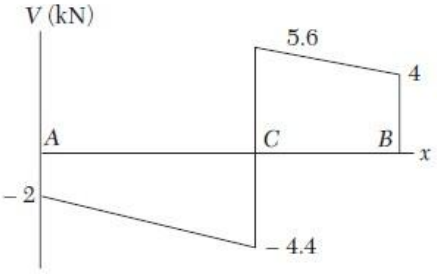
$$(0.6)(2) + (0.1)(4) + (0.4)B = 0$$

$$B = -4 \text{ kN}$$

$$\circlearrowleft \Sigma M_B = 0$$

$$(1.0)(2) + (0.5)(4) - (0.4)C = 0$$

$$C = 10 \text{ kN}$$



A to C $\quad 0 < x < 0.6$ m

$$+\uparrow \Sigma F_y = 0$$

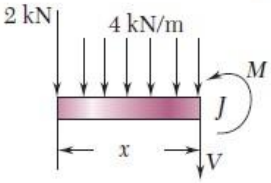
$$-2 - 4x - V = 0$$

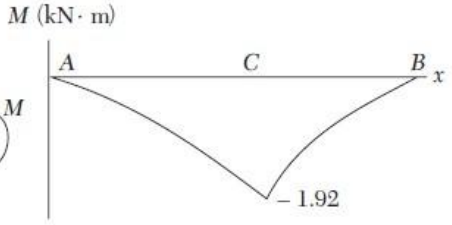
$$V = -2 - 4x$$

$$\circlearrowleft \Sigma M_J = 0$$

$$2x + (4x)\left(\frac{x}{2}\right) + M = 0$$

$$M = -2x^2 - 2x$$





At C $\quad M = 1.92$ kN · m

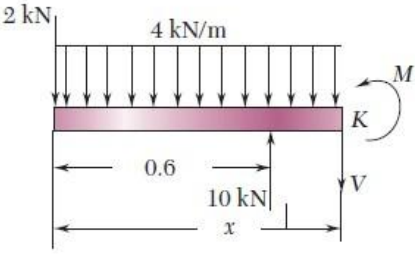
C to B $\quad 0.6 \text{ m} < x < 1.0$ m

$$+\uparrow \Sigma F_y = 0 \qquad -2 - 4x + 10 - V = 0 \qquad V = 8 - 4x$$

$$\circlearrowleft \Sigma M_K = 0 \qquad -2x + (4x)\left(\frac{x}{2}\right) - (10)(x - 0.6) + M = 0$$

$$M = -2x^2 + 8x - 6$$

$$\max |M| = 1.92 \text{ kN} \cdot \text{m} = 1.92 \times 10^3 \text{ N} \cdot \text{m}$$



For rolled steel section $\quad S\,100 \times 11.5 \qquad S = 49.6 \times 10^3 \text{ mm}^3$

$$= 49.6 \times 10^{-6} \text{ m}^3$$

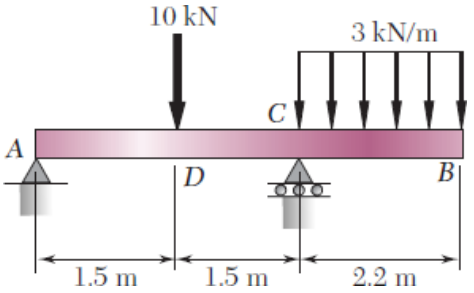
Maximum normal stress $\quad \sigma = \dfrac{|M|}{S} = \dfrac{1.92 \times 10^3}{49.6 \times 10^{-6}} = 38.7 \times 10^6 \text{ Pa}$

$$= 38.7 \text{ MPa}$$

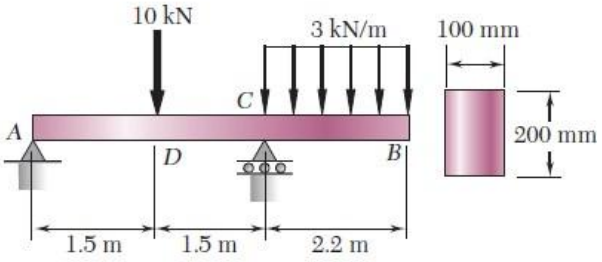
◄

Question 6.

For the beam and loading shown, draw the Shear force and bending moment diagrams by inspection and determine the bending moment on a transverse section at C.



SOLUTION



$$\circlearrowleft \Sigma M_C = 0$$

$$-3A + (1.5)(10) - (1.1)(2.2)(3) = 0$$

$$A = 2.58 \text{ kN}$$

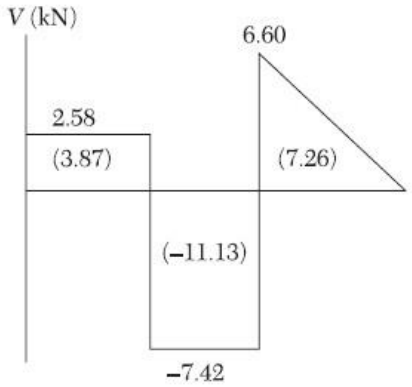
$$\circlearrowleft \Sigma M_A = 0$$

$$-(1.5)(10) + 3C - (4.1)(2.2)(3) = 0$$

$$C = 14.02 \text{ kN}$$

Shear

A to D^-	$V = 2.58$ kN
D^+ to C^-	$V = 2.58 - 10 = -7.42$ kN
C^+	$V = -7.42 + 14.02 = 6.60$ kN
B	$V = 6.60 - (2.2)(3) = 0$

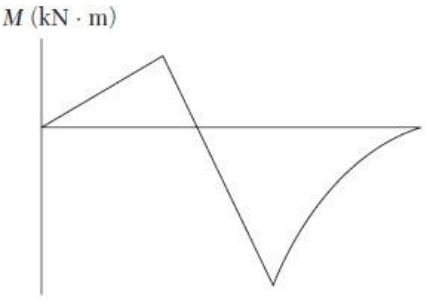


Areas under shear diagram

A to D $\quad \int V\,dx = (1.5)(2.58) = 3.87 \text{ kN} \cdot \text{m}$

D to C $\quad \int V\,dx = (1.5)(-7.42) = -11.13 \text{ kN} \cdot \text{m}$

C to B $\quad \int V\,dx = \left(\frac{1}{2}\right)(2.2)(6.60) = 7.26 \text{ kN} \cdot \text{m}$



Bending moments

$$M_A = 0$$

$$M_D = 0 + 3.87 = 3.87 \text{ kN} \cdot \text{m}$$

$$M_C = 3.87 - 11.13 = -7.26 \text{ kN} \cdot \text{m}$$

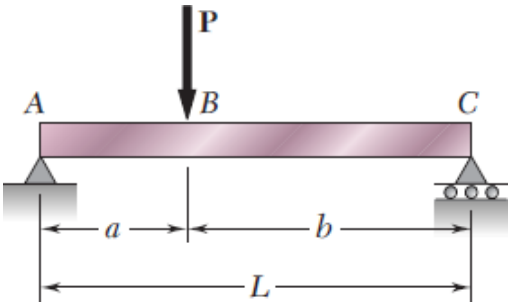
$$M_B = 7.26 - 7.26 = 0$$

$|M_C| = 7.26 \text{ kN} \cdot \text{m} = 7.26 \cdot 10^3 \text{ N} \cdot \text{m}$

Question 7.

For the beam and loading shown, draw the Shear force and bending moment diagrams by inspection and determine the maximum bending moment.

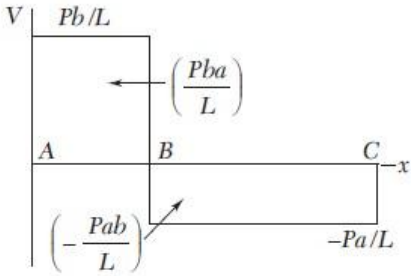
Use P = 4kN, a = 2m, b=3m



SOLUTION

$$\circlearrowleft^{+} \Sigma M_C = 0 \qquad LA - bP = 0 \qquad A = \frac{Pb}{L}$$

$$\circlearrowleft^{+} \Sigma M_A = 0 \qquad LC - aP = 0 \qquad C = \frac{Pa}{L}$$



At A^+ $\qquad V = A = \frac{Pb}{L} \qquad M = 0$

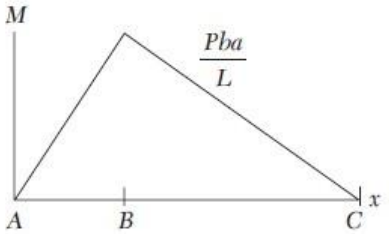
A to B^- $\qquad 0 < x < a$

$$w = 0 \qquad \int_o^x w\,dx = 0$$

$$V_B - V_A = 0 \qquad V_B = \frac{Pb}{L}$$

$$M_B - M_A = \int_o^a V dx = \int_o^a \frac{Pb}{L}\,dx = \frac{Pba}{L}$$

$$M_B = \frac{Pba}{L}$$



At B^+ $\qquad V = A - P = \frac{Pb}{L} - P = -\frac{Pa}{L}$

B^+ to C $\qquad a < x < L$

$$w = 0 \qquad \int_a^x w\,dx = 0$$

$$V_C - V_B = 0 \qquad V = -\frac{Pa}{L}$$

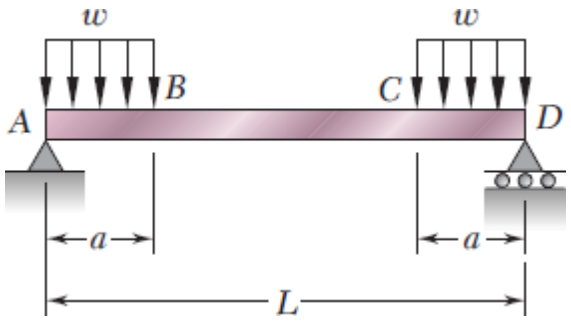
$$M_C - M_B = \int_a^L V\,dx = -\frac{Pa}{L}(L - a) = \frac{Pab}{L}$$

$$M_C = M_B - \frac{Pab}{L} = \frac{Pba}{L} - \frac{Pab}{L} = 0$$

Question 8.

For the beam and loading shown, draw the Shear force and bending moment diagrams by inspection and determine the maximum bending moment.

Use ω = 4kN/m, a=1.5m, L=6m



SOLUTION

Reactions $A = D = wa$

A to B $\quad 0 < x < a \quad w = w$

$$V_A = A = wa, \qquad M_A = 0$$

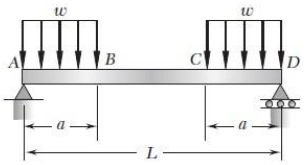
$$V - V_A = \int_o^x w\, dx = -wx$$

$$V = w(a - x) \qquad V_B = 0$$

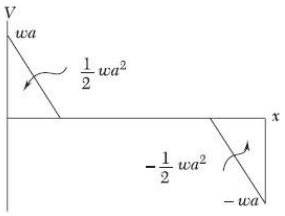
$$\frac{dM}{dx} = V = wa - wx$$

$$M - M_A = \int_a^x V\, dx = \int_o^x (wa - wx)\, dx$$

$$= wax - \frac{1}{2}wx^2$$



$$M_B = \frac{1}{2}\, wa^2 \text{ at } x = a$$

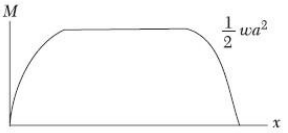


B to C $\quad a < x < L - a \qquad V = 0$

$$\frac{dM}{dx} = V = 0$$

$$M - M_B = \int_a^x v\, dx = 0$$

$$M = M_B = \frac{1}{2}wa^2$$



C to D $\quad V - V_C = -\int_{L-a}^{x} w\, dx = -w\,[x - (L - a)]$

$$V = -w\,[x - (L - a)]$$

$$M - M_C = \int_{L-a}^{x} V\, dx = \int_{L-a}^{x} -w\,[x - (L - a)]dx$$

$$= -w\left[\frac{x^2}{2} - (L - a)x\right]\Bigg|_{L-a}^{x}$$

$$= -w\left[\frac{x^2}{2} - (L - a)x - \frac{(L - a)^2}{2} + (L - a)^2\right]$$

$$= -w\left[\frac{x^2}{2} - (L - a)x + \frac{(L - a)^2}{2}\right]$$

$$M = \frac{1}{2}\, wa^2 - w\left[\frac{x^2}{2} - (L - a)x + \frac{(L - a)^2}{2}\right]$$